

Лабораторная работа №14

Настройка файловых служб Samba

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06 декабря 2025

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Цель работы

Приобрести навыки настройки доступа групп пользователей к общим ресурсам по протоколу **SMB/Samba**.

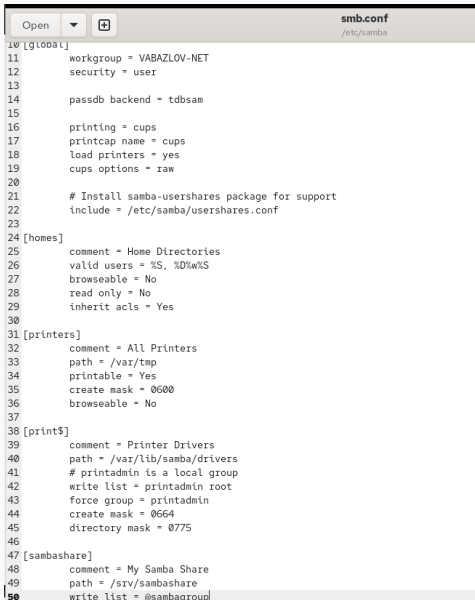
Настройка сервера Samba

```
Installed:
  cifs-utils-7.2-1.el10.x86_64
  samba-4.22.4-106.el10.x86_64
  samba-common-tools-4.22.4-106.el10.x86_64
  samba-ldb-ldap-modules-4.22.4-106.el10.x86_64
  libnetapi-4.22.4-106.el10.x86_64
  samba-client-4.22.4-106.el10.x86_64
  samba-dcerpc-4.22.4-106.el10.x86_64
  samba-libs-4.22.4-106.el10.x86_64

Complete!
[root@server.vabazlov.net server]#
[root@server.vabazlov.net server]# groupadd -g 1010 sambagroup
[root@server.vabazlov.net server]# usermod -aG sambagroup vabazlov
[root@server.vabazlov.net server]# mkdir -p /srv/smbashare
[root@server.vabazlov.net server]#
```

Рис. 1: Установка пакетов

Конфигурация smb.conf

A screenshot of a text editor window titled 'smb.conf' with a subtitle '/etc/samba'. The editor has a menu bar with 'Open' and a '+' icon. The content is a configuration file for Samba, showing sections for global settings, home directories, printers, and a share named 'sambashare'. Line numbers 10 through 50 are visible on the left margin.

```
10 [global]
11     workgroup = VABAZLOV-NET
12     security = user
13
14     passdb backend = tdbsam
15
16     printing = cups
17     printcap name = cups
18     load printers = yes
19     cups options = raw
20
21     # Install samba-usershares package for support
22     include = /etc/samba/usershares.conf
23
24 [homes]
25     comment = Home Directories
26     valid users = %S, %D%w%S
27     browseable = No
28     read only = No
29     inherit acls = Yes
30
31 [printers]
32     comment = All Printers
33     path = /var/tmp
34     printable = Yes
35     create mask = 0600
36     browseable = No
37
38 [print$]
39     comment = Printer Drivers
40     path = /var/lib/samba/drivers
41     # printadmin is a local group
42     write list = printadmin root
43     force group = printadmin
44     create mask = 0664
45     directory mask = 0775
46
47 [sambashare]
48     comment = My Samba Share
49     path = /srv/sambashare
50     write list = @sambaagroup
```

Проверка и запуск сервиса

```
[root@server.vabazlov.net server]# systemctl start smb
[root@server.vabazlov.net server]# systemctl enable smb
Created symlink '/etc/systemd/system/multi-user.target.wants/smb.service' → '/usr/lib/systemd/system/smb.service'.
[root@server.vabazlov.net server]# systemctl status smb
● smb.service - Samba SMB Daemon
   Loaded: loaded (/usr/lib/systemd/system/smb.service; enabled; preset: disabled)
   Active: active (running) since Sat 2025-12-06 10:26:28 UTC; 17s ago
 Invocation: d804f959eb604b1c8b780988aea958ba
    Docs: man:smbd(8)
          man:samba(7)
          man:smb.conf(5)
 Main PID: 16573 (smbd)
   Status: "smbd: ready to serve connections..."
    Tasks: 3 (limit: 10381)
  Memory: 13.3M (peak: 13.6M)
     CPU: 31ms
   CGroup: /system.slice/smb.service
           └─16573 /usr/sbin/smbd --foreground --no-process-group
             └─16576 /usr/sbin/smbd --foreground --no-process-group
               └─16577 /usr/sbin/smbd --foreground --no-process-group

Dec 06 10:26:27 server.vabazlov.net systemd[1]: Starting smb.service - Samba SMB Daemon...
Dec 06 10:26:28 server.vabazlov.net systemd[1]: Started smb.service - Samba SMB Daemon.
[root@server.vabazlov.net server]# smbclient -L //server
Password for [VABAZLOV-NET\root]:
Anonymous login successful

      Sharename      Type            Comment
      -----
      print$         Disk            Printer Drivers
      sambashare      Disk            My Samba Share
      IPC$            IPC             IPC Service (Samba 4.22.4)

SMB1 disabled -- no workgroup available
[root@server.vabazlov.net server]#
```

Рис. 3: Доступ через smbclient

```
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>Samba</short>
  <description>This option allows you to access and participate in Windows file and printer sharing networks. You need the samba package installed for this option to be useful.</description>
  <include service="samba-client"/>
  <port protocol="tcp" port="139"/>
  <port protocol="tcp" port="445"/>
</service>
/usr/lib/firewalld/services/samba.xml (END)
```

Рис. 4: Описание сервиса Samba


```
[root@server.vabazlov.net server]#  
[root@server.vabazlov.net server]# firewall-cmd --add-service=samba  
success  
[root@server.vabazlov.net server]# firewall-cmd --add-service=samba --permanent  
success  
[root@server.vabazlov.net server]# firewall-cmd --reload  
success  
[root@server.vabazlov.net server]# chgrp sambagroup /srv/sambashare/  
[root@server.vabazlov.net server]# chmod g=rwx /srv/sambashare/  
[root@server.vabazlov.net server]# cd /srv/  
[root@server.vabazlov.net srv]# ls -Z  
unconfined_u:object_r:nfs_t:s0 nfs unconfined_u:object_r:var_t:s0 sambashare  
[root@server.vabazlov.net srv]# semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"  
[root@server.vabazlov.net srv]# ls -Z  
unconfined_u:object_r:nfs_t:s0 nfs unconfined_u:object_r:var_t:s0 sambashare  
[root@server.vabazlov.net srv]# restorecon -vR /srv/sambashare/  
Relabeled /srv/sambashare from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:samba_share_t:s0  
[root@server.vabazlov.net srv]# ls -Z  
unconfined_u:object_r:nfs_t:s0 nfs unconfined_u:object_r:samba_share_t:s0 sambashare  
[root@server.vabazlov.net srv]# setsebool samba_export_all_rw 1  
[root@server.vabazlov.net srv]# setsebool samba_export_all_rw 1 -P  
[root@server.vabazlov.net srv]# smbpasswd -L -a vabazlov  
New SMB password:  
Retype new SMB password:  
Added user vabazlov.  
[root@server.vabazlov.net srv]# █
```

Рис. 5: Настройка прав доступа

```
[vabazlov@server.vabazlov.net common]$  
[vabazlov@server.vabazlov.net common]$ id  
uid=1001(vabazlov) gid=1001(vabazlov) groups=1001(vabazlov),10(wheel) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023  
[vabazlov@server.vabazlov.net common]$ newgrp sambagroup  
[vabazlov@server.vabazlov.net common]$ id  
uid=1001(vabazlov) gid=1010(sambagroup) groups=1010(sambagroup),10(wheel),1001(vabazlov) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023  
[vabazlov@server.vabazlov.net common]$ cd /srv/smbashare/  
[vabazlov@server.vabazlov.net sambashare]$ ls  
[vabazlov@server.vabazlov.net sambashare]$ touch vabazlov@server.txt  
[vabazlov@server.vabazlov.net sambashare]$ ls  
vabazlov@server.txt  
[vabazlov@server.vabazlov.net sambashare]$  
[vabazlov@server.vabazlov.net sambashare]$
```

Рис. 6: Проверка групп

Монтирование Samba на клиенте

```
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>Samba Client</short>
  <description>This option allows you to access Windows file and printer sharing networks. You need the samba-client package installed for this option to be useful.</description>
  <include service="netbios-ns"/>
  <port protocol="udp" port="138"/>
</service>
/usr/lib/firewalld/services/samba-client.xml (END)
```

Рис. 7: Описание samba-client

```
Installed:
  cifs-utils-7.2-1.el10.x86_64          samba-client-4.22.4-106.el10.x86_64

Complete!
[root@client.vabazlov.net client]#
[root@client.vabazlov.net client]# less /usr/lib/firewalld/services/samba-client.xml
[root@client.vabazlov.net client]#
[root@client.vabazlov.net client]#
[root@client.vabazlov.net client]# firewall-cmd --add-service=samba-client
success
[root@client.vabazlov.net client]# firewall-cmd --add-service=samba-client --permanent
success
[root@client.vabazlov.net client]# firewall-cmd --reload
success
[root@client.vabazlov.net client]# groupadd -g 1010 sambagroup
[root@client.vabazlov.net client]# usermod -aG sambagroup vabazlov
[root@client.vabazlov.net client]#
```

Рис. 8: Создание группы sambagroup

Проверка доступных ресурсов

```
[root@client.vabazlov.net client]#  
[root@client.vabazlov.net client]# smbclient -L //server  
Password for [VABAZLOV-NET\root]:  
Anonymous login successful
```

Sharename	Type	Comment
-----	----	-----
print\$	Disk	Printer Drivers
smbashare	Disk	My Samba Share
IPC\$	IPC	IPC Service (Samba 4.22.4)

SMB1 disabled -- no workgroup available

```
[root@client.vabazlov.net client]# smbclient -L //server -U vabazlov  
Password for [VABAZLOV-NET\vabazlov]:
```

Sharename	Type	Comment
-----	----	-----
print\$	Disk	Printer Drivers
smbashare	Disk	My Samba Share
IPC\$	IPC	IPC Service (Samba 4.22.4)
vabazlov	Disk	Home Directories

SMB1 disabled -- no workgroup available

```
[root@client.vabazlov.net client]# █
```

Рис. 9: Просмотр ресурсов

```
[root@client.vabazlov.net client]#  
[root@client.vabazlov.net client]# mkdir /mnt/samba  
[root@client.vabazlov.net client]# mount -o username=vabazlov,user,rw,uid=vabazlov,gid=sambagroup //server/sambashare /mnt/samba/  
Password for vabazlov@//server/sambashare:  
mount: (hint) your fstab has been modified, but systemd still uses  
the old version; use 'systemctl daemon-reload' to reload.  
[root@client.vabazlov.net client]# systemctl daemon-reload  
[root@client.vabazlov.net client]# mount | grep samba  
//server/sambashare on /mnt/samba type cifs (rw,nosuid,nodev,noexec,relatime,vers=3.1.1  
,cache=strict,upcall_target=app,username=vabazlov,uid=1001,forceuid,gid=1010,forcegid,a  
ddr=192.168.1.1,file_mode=0755,dir_mode=0755,soft,nounix,serverino,mapposix,repase=nfs  
,rsize=4194304,wsz=4194304,bsize=1048576,retrans=1,echo_interval=60,actimeo=1,closeti  
meo=1,user)  
[root@client.vabazlov.net client]#
```

Рис. 10: Монтирование ресурса

```
[vabazlov@server.vabazlov.net common]$ cd /srv/smbshare/
[vabazlov@server.vabazlov.net sambashare]$ ls
[vabazlov@server.vabazlov.net sambashare]$ touch vabazlov@server.txt
[vabazlov@server.vabazlov.net sambashare]$ ls
vabazlov@server.txt
[vabazlov@server.vabazlov.net sambashare]$
[vabazlov@server.vabazlov.net sambashare]$
[vabazlov@server.vabazlov.net sambashare]$ ls -l
total 0
-rwxr--r--. 1 vabazlov vabazlov  0 Dec  6 10:42 vabazlov@client.txt
-rw-r--r--. 1 vabazlov sambagroup 0 Dec  6 10:31 vabazlov@server.txt
[vabazlov@server.vabazlov.net sambashare]$ █
```

Рис. 11: Проверка файла



```
1
2 #
3 # /etc/fstab
4 # Created by anaconda on Wed Sep 3 08:53:23 2025
5 #
6 # Accessible filesystems, by reference, are maintained under '/dev/disk/'.
7 # See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
8 #
9 # After editing this file, run 'systemctl daemon-reload' to update systemd
10 # units generated from this file.
11 #
12 UUID=908334c5-81ab-40aa-ad89-1bd296275208 / xfs defaults 0 0
13 UUID=964b6a41-d718-411a-a362-f90f740c04d8 /boot xfs defaults 0 0
14 UUID=2000-E25B /boot/efi vfat umask=0077,shortname=winnt 0 2
15 UUID=7e0ea0d4-0dbc-4e9b-bcd8-a0b4c565de96 /home xfs defaults 0 0
16 UUID=3c01d968-53fe-4cb3-8a7c-cc0f286d9e58 none swap defaults 0 0
17 server.vabazlov.net:/srv/nfs /mnt/nfs nfs _netdev 0 0
18 /server/smbashare /mnt/samba cifs user,rw,uid=vabazlov,gid=sambagroup,credentials=/etc/samba/
   smbusers,_netdev 0 0
19 #VAGRANT-BEGIN
20 # The contents below are automatically generated by Vagrant. Do not modify.
21 vagrant /vagrant vboxsf uid=1000,gid=1000,_netdev 0 0
22 #VAGRANT-END
```

Рис. 12: fstab для SMB

Итоги работы

- Настроены сервер и клиент Samba
- Реализована групповая модель доступа
- Проверены SELinux и межсетевой экран
- Настроено монтирование вручную и через fstab
- Созданы автоматизирующие скрипты для обеих ВМ